

(Salary, age)

- Classification: # 0 - 50 ~~th~~
1 - 500

Accuracy - score also does wrong.
102. ^{Calw pps says 17} dumb mode also gives higher accuracy.

Majority $\rightarrow$ do ^{random} sampling $\rightarrow$ take some no. of majority rows same as minority.

- from imblearn. under-sampling import RandomUnderSampler.

0 - 50

1 - 50

- Disadvantage - lose of data.

Q3

Q3 Over Sampling.

#maj: 900

#min: 100

increase to 900 by doing duplication.

from imblearn. Over-sampling import RandomOverSamp

• It can be seen by scatterplot as of points overlapping.

Disadv: overfitting. } Don't do with Decision Trees.

~~SMOTE~~ **SMOTE** [Synthetic minority over sampling technique]
generate new random dataset to increase the #minority.
[betⁿ 2 minority points].

Steps:

$K \sim 5$

- Train KNN on minority - find each observation's closest points.

To create new synthetic data:

- Select any point.
- Find the neighbour point.
- Calc:

$$\text{new point} = \text{Original Sample} - \text{factor}^* (\text{original point} - \text{neighbour point})$$

$$= (1, 2) - 0.5 * ((1, 1) - (2, 2))$$

$$= (2.5, 1.5)$$

• add.

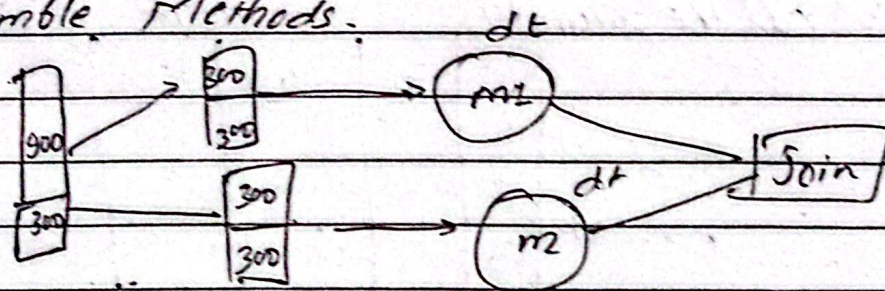
from imblearn, over-sampling import SMOTE.

Disadvantages:

- not for categorical data
- Sensitive to outliers.

4

Ensemble Methods:



- Balanced Random Forest Classifier.

5

Cost Sensitive Learning:

weight in class. [Logistic Reg, SVM, DT]
[not in KNN]

Loss Function